

- 1 Clusters can be formed by “cutting” the dendrogram at a user-chosen
- 2 height. It may seem “clear” by looking at the dendrogram that cutting at
- 3 around a height of 3 would be a good choice. The results is three clusters.
- 4 Note that in this case I generated the ten observations from the normal
- 5 distribution with a standard deviation of 0.5. The mean for observations 1
- 6 thru 5 was 0, for observations 6 thru 8 was 5, and for observations 9 and
- 7 10 was 7. This explains the structure in the dendrogram.

1 **Building the Dendrogram**

- 2 The process of constructing the dendrogram is quite intuitive.
- 3 First, the two observations which are the most similar (as measured by the
- 4 chosen dissimilarity metric) and joined together, at a height equal to their
- 5 dissimilarity. In our example above, observations 1 and 4 were the first to
- 6 be joined, at a height very close to zero.
- 7 This process then continues, with the most similar objects joined together
- 8 at the appropriate height. **But:** Once observations are joined, they are
- 9 **treated as a unit** from that point on. For instance, in the second step in
- 10 the construction of the dendrogram above, observation 3 was joined with
- 11 the **unit consisting of 1 and 4**.
- 12 This process continues until all observations are joined.

1 The obvious question is thus: **How is the dissimilarity of collections of**
2 **observations calculated?** There are a few different approaches. Two of the
3 standard ones are as follows:

4 With **complete linkage**, the distance between sets of observations, S_i and
5 S_j , is defined to be the **maximal** dissimilarity between all pairs $x \in S_i$ and
6 $y \in S_j$. The default in R `hclust()`

7 With **single linkage**, the distance between sets of observations, S_i and S_j ,
8 is defined to be the **minimal** dissimilarity between all pairs $x \in S_i$ and
9 $y \in S_j$.

10 Less-commonly used options are **centroid linkage**, in which the distance
11 between S_i and S_j is calculated as the dissimilarity between the centroids
12 of the two groups, and **average linkage**, calculated as the average distance
13 between all pairs of elements in S_i and S_j .

1 Hierarchical Clustering in R

2 The function `hclust()` performs hierarchical clustering in R.

3 Remember that this method is built around a user-chosen measure of dis-
4 similarity. Hence, the primary input argument to `hclust()` is a symmet-
5 ric n by n distance matrix. The diagonal of this matrix should consist of
6 zeros.

- The syntax for complete linkage clustering is as follows:

```
> hcout = hclust(dist(t(stocksamplescl[,1:100])),
+               method="complete")
```

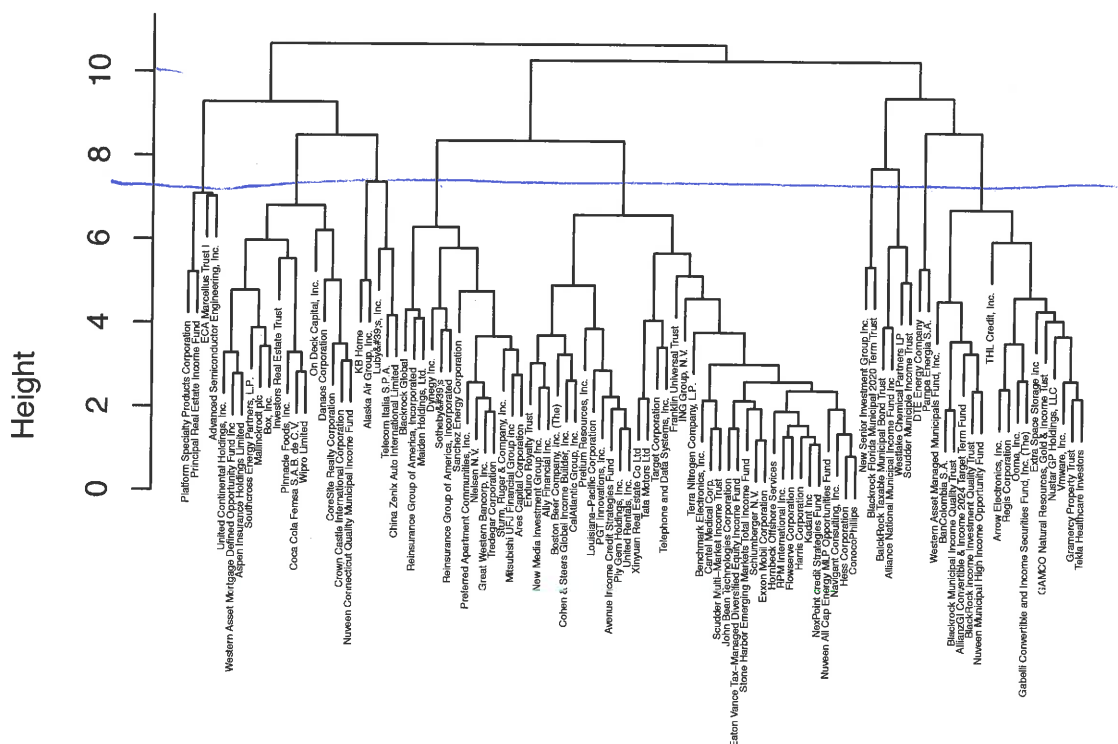
- (Note that only the first 100 stocks are used, since the dendrogram becomes difficult to read otherwise. There is no such practical limit in using this method, however.)

- 5 Other choices for method include single, centroid, and average.

- 6 The `plot()` function, when applied to the output of `hclust()`, produces
7 the dendrogram:

```
> plot(hcout, labels=stocksample$name[1:100],
+      cex=0.35, sub="", xlab="")
```

Cluster Dendrogram



- 1 Another useful function is `cutree()`. This will create return cluster membership for each of the observation, based on a desired number of clusters,
- 2 or on a desired height.

- 4 For instance, to cut our dendrogram into six clusters, we specify `k=6`:

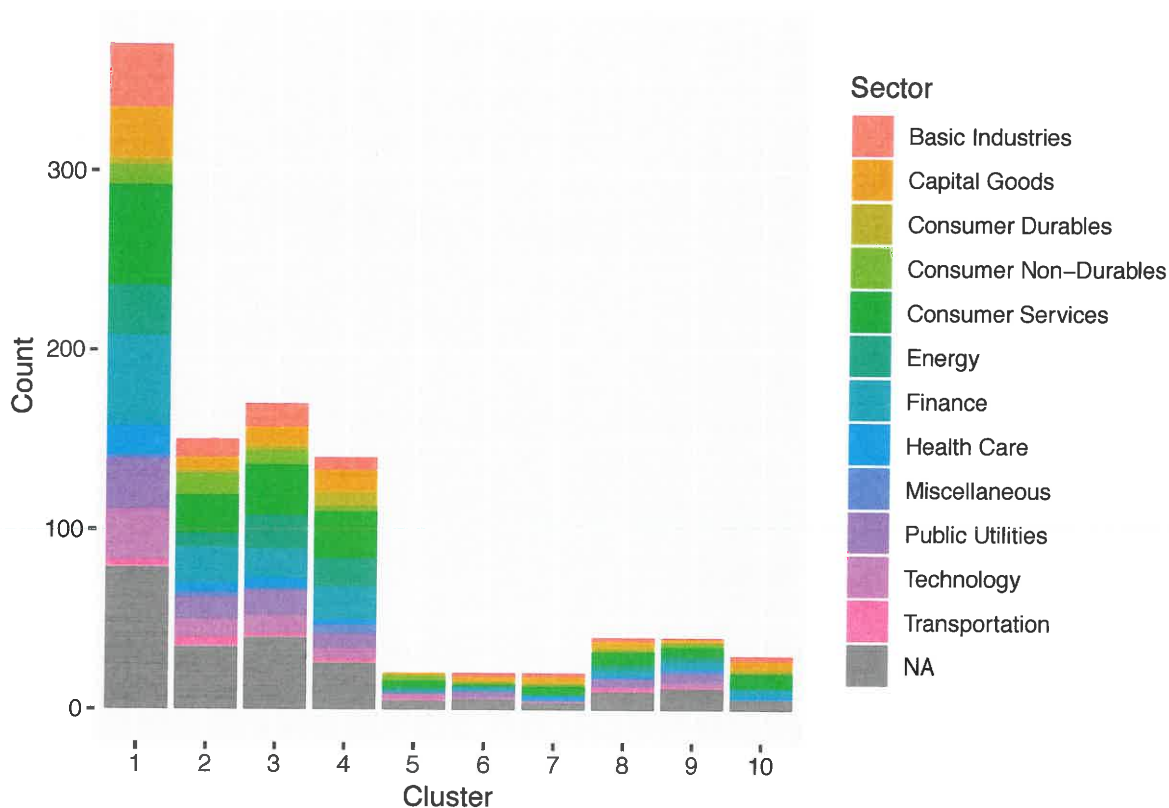
```
> cutree(hcout, k=6) [1:10]
[1] 1 2 3 2 2 1 1 1 1 2 , . -
```

- 5 And to cut our our dendrogram at a height of 10, specify `h=10`:

```
> cutree(hcout, h=10) [1:10]
[1] 1 2 3 2 2 1 1 1 1 2
```

- 1 We can again inspect how the clusters vary with respect to sector:

```
> stocksample$hcclust = factor(cutree(hcout, 10))
>
> ggplot(stocksample, aes(x=hcclust, fill=sector)) +
+   geom_bar() +
+   labs(x="Cluster", y="Count", fill="Sector")
```



smooth structure

$$Y_{i1} = f_{i1} + \epsilon_{i1} \quad i=1, 2, \dots, 30$$

observed

noise, mean 0

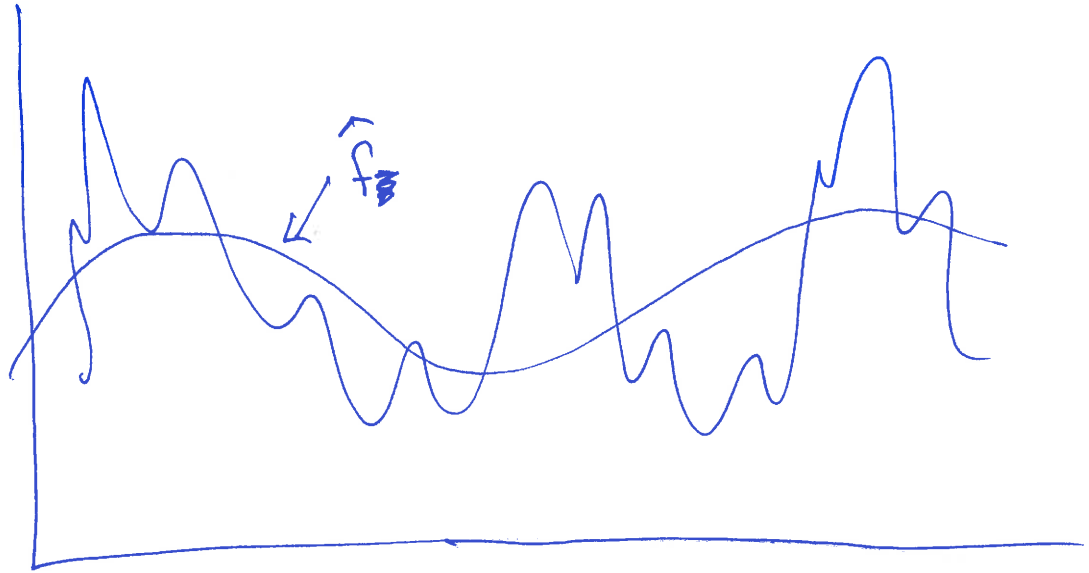
$$Y_{i2} = f_{i2} + \epsilon_{i2}$$

$$\sum_{i=1}^{30} (Y_{i1} - Y_{i2})^2 = \sum_{i=1}^{30} (f_{i1} + \epsilon_{i1} - f_{i2} - \epsilon_{i2})^2$$

$$= \underbrace{\sum_{i=1}^{30} (f_{i1} - f_{i2})^2}_{\text{distance of interest}} + \underbrace{\sum_{i=1}^{30} (\epsilon_{i1} - \epsilon_{i2})^2}_{\text{each has variance } 2\sigma^2, \text{ if } \sigma^2 = V(\epsilon_{ij})} + \text{cross products (small)}$$

What can we do?

Start by smoothing the time series, using
local linear regression or another nonparametric



Run clustering on $\hat{f}'_T$'s

Or HW

1 Part 10: Fitting Parametric Models

2 Summary

3 In this Part we begin consideration of the problem of fitting parametric
4 probability models to observed data.

5 Typically, a probability model is **incompletely specified**. I.e., a particular
6 model is assumed, but there are still unknown components, often real-
7 valued **parameters**. We use available data (a **sample**) to estimate these
8 parameters.

1 **Example:** In a certain situation, an individual is comfortable assuming that
2 $X_1, X_2, \dots, X_n$ are **independent and identically distributed (i.i.d.)** with the
3 Normal distribution with mean μ and variance σ^2 . These parameters are
4 unknown, and must be estimated from data.

5 Here, we will propose **estimators** for μ and σ^2 and explore their properties.

6 How, in general, are estimators constructed? And, how is the uncertainty
7 in these estimators quantified?

1 Terminology

2 The **parameters** are the constants that distinguish probability models in the
3 same "family." Examples of "families" include "normal," "exponential,"
4 and so forth.

5 A **statistic** is a function of the RVs being used to model the observable data.
6 Examples include the sample mean, the sample variance, etc. A statistic is
7 a **random variable**. $X_1, X_2, \dots, X_n$ RV's $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

8 An **estimator** is a statistic which is used to estimate an unknown parameter.
9 It is also a **random variable**.

10 The "hat notation" is standard for denoting an estimator: $\hat{\beta}$ is an estimator
11 for β , for example. θ is often used to denote a generic parameter, $\hat{\theta}$ is its
12 estimator.

1 **Exercise:** Explain the sense in which an estimator is appropriately thought
2 of as a random variable.

3 $X_1, X_2, \dots, X_n$ are RVs that model the
4 observable data, i.e. we imagine we have
5 observed "draws" from these distributions

6 The estimator $\hat{\Theta}$ for θ is a function of
7 $X_1, X_2, \dots, X_n$, i.e. $\hat{\Theta} = f(X_1, X_2, \dots, X_n)$

8 $\hat{\Theta}(\omega) = f(X_1(\omega), X_2(\omega), \dots, X_n(\omega))$,
9
10 i.e. $\hat{\Theta}$ is a function on Ω , a RV
11

1 Performance of Estimators

2 Our objective is to estimate parameters, i.e., to construct estimators. For
3 any situation, there are many choices of estimator, so we require methods
4 to assess the performance of an estimator.

5 The “performance” is closely tied to the distribution of the estimator. The
6 distributions we learned in Probability (plus some more) are useful for this
7 purpose.

8 An ideal estimator $\hat{\theta}$ for θ would have two properties:

- 9 1. The estimator should be **accurate**, i.e., its distribution would be cen-
10 tered on the true value θ .
- 11 2. The estimator should be **precise**, i.e., its distribution has a small spread
12 (small variance)

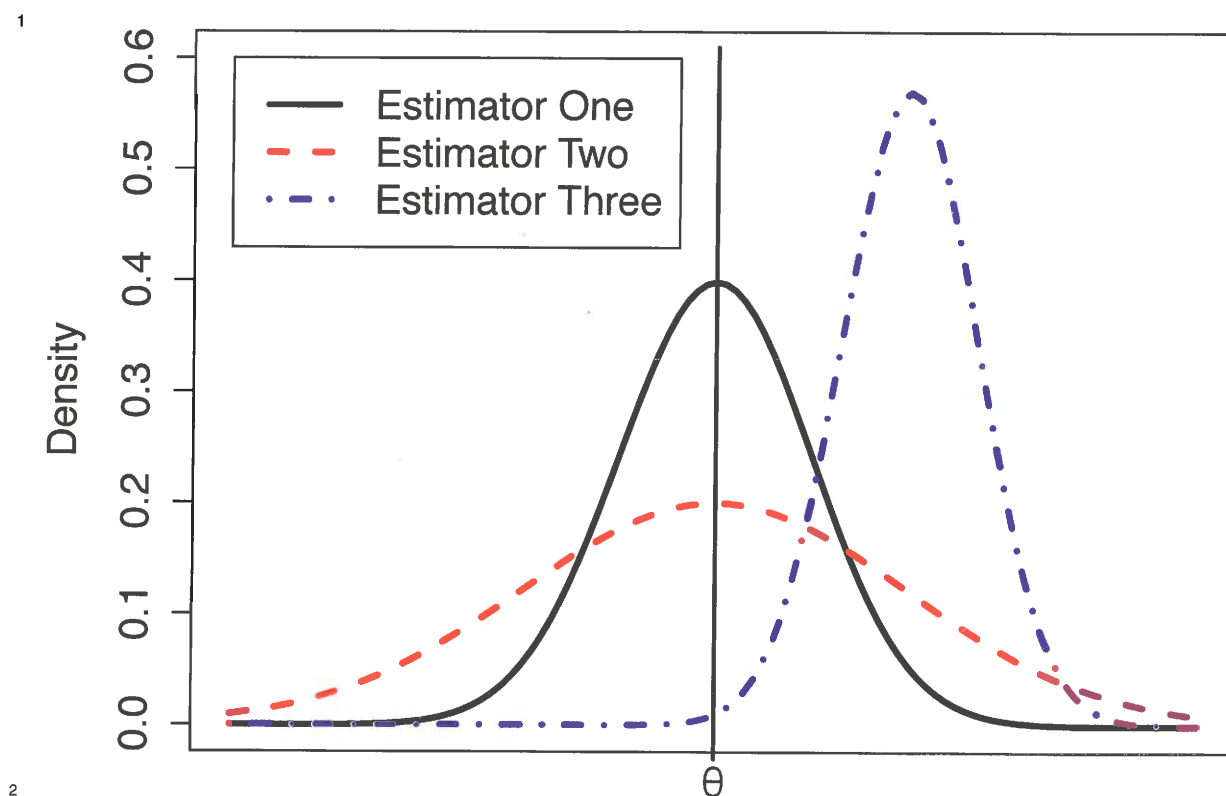
1 **Example:** The figure on the following page compares the distributions of
2 three hypothetical estimators for a parameter θ , call them $\hat{\theta}_1$, $\hat{\theta}_2$, and $\hat{\theta}_3$.

3 We note that $E(\hat{\theta}_1) = \theta$, so it is optimally accurate. Also, $E(\hat{\theta}_2) = \theta$.

4 But, $V(\hat{\theta}_1) < V(\hat{\theta}_2)$, so $\hat{\theta}_1$ is more precise than $\hat{\theta}_2$.

5 So, we would prefer $\hat{\theta}_1$ to $\hat{\theta}_2$.

6 Also, $V(\hat{\theta}_3) < V(\hat{\theta}_1)$, but $\hat{\theta}_3$ has terrible accuracy.



- 1 An estimator $\hat{\theta}$ is said to be **an unbiased estimator for θ** if $E(\hat{\theta}) = \theta$.
optimally accurate
- 2 The variability in an estimator (its precision) is naturally quantified by its
- 3 variance, $V(\hat{\theta})$.
- 4 The **standard error (SE) of $\hat{\theta}$** is the square root of its variance. The SE is of-
- 5 ten reported along with an estimate in order to give a sense of the amount
- 6 of error one should anticipate in that estimate.
- 7 An estimator $\hat{\theta}$ is said to be the **minimum variance unbiased estimator for**
- 8 **θ (MVUE)** if $\hat{\theta}$ is unbiased for θ and for any other unbiased estimator $\hat{\theta}'$,
- 9 $V(\hat{\theta}) \leq V(\hat{\theta}')$.

- 1 **Exercise:** Suppose that X has the binomial(n, p) distribution. It is com-
 2 mon to estimate p using $\hat{p} = X/n$, the sample proportion. Is this estimator
 3 unbiased? What is its variance and standard error?

4 To check unbiased:

$$5 \quad E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = p$$

6 So, $\hat{p}$ is unbiased for p .

$$9 \quad V(\hat{p}) = V\left(\frac{X}{n}\right) = \frac{1}{n^2} V(X) = \frac{p(1-p)}{n}$$

$$11 \quad \text{So, } SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

12 Usually, approx. using $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

- 1 There is often a “tradeoff” between bias and variance (accuracy and preci-
 2 sion). This is called the **bias-variance tradeoff**.
 3 In some cases, allowing some bias leads to reduced variance.

- 4 This tradeoff is made formal via the **mean squared error (MSE)**. For an
 5 estimator $\hat{\theta}$ for θ ,

$$6 \quad \text{MSE}_{\theta}(\hat{\theta}) = E\left((\hat{\theta} - \theta)^2\right)$$

- 7 i.e., it is the expected squared error in estimating θ when using $\hat{\theta}$.

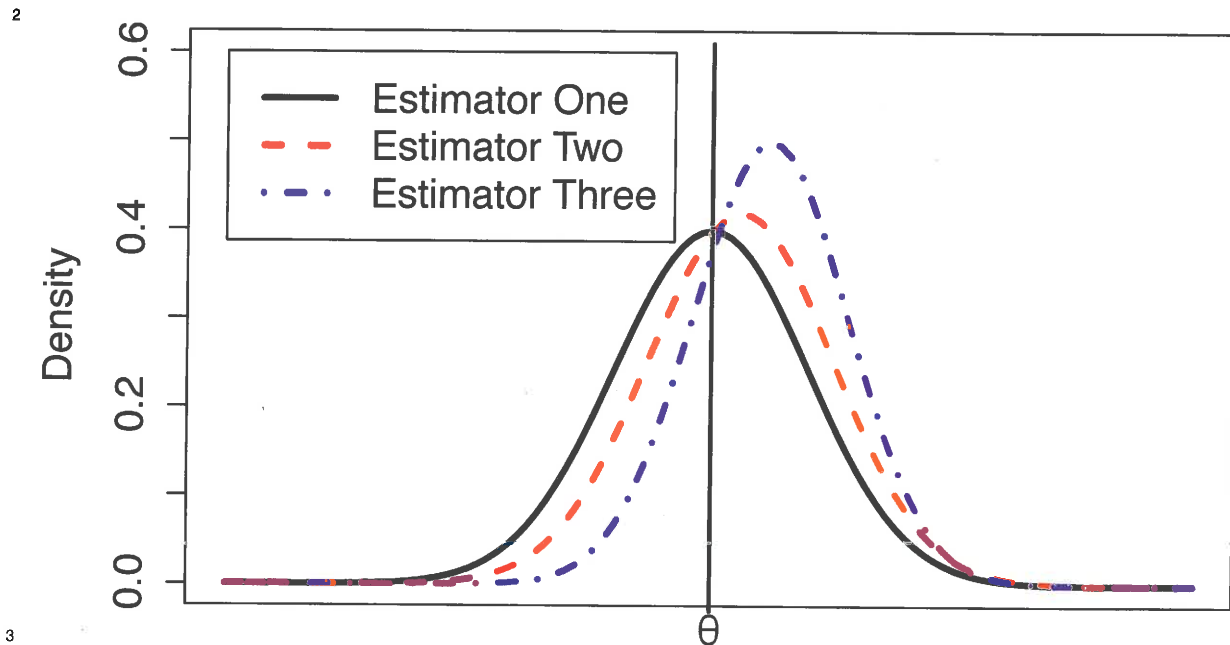
- 8 One can show that

$$9 \quad \text{MSE}_{\theta}(\hat{\theta}) = V(\hat{\theta}) + \text{bias}_{\theta}^2(\hat{\theta})$$

- 10 where

$$11 \quad \text{bias}_{\theta}(\hat{\theta}) = E(\hat{\theta}) - \theta, \text{ i.e. if unbiased, then bias} = 0$$

- 1 In the example below, all three estimators have the same MSE.



1 Constructing Estimators: An Example

- 2 The binomial example considered above is misleading in one respect: In
3 that case it is “obvious” what the natural estimator is, since one can use
4 the sample proportion to estimate the population proportion.

- 5 In other cases, the choice of estimator is not so clear.

- 6 We first consider a simple approach to constructing estimators, the **method**
7 **of moments**.

1 First, recall that for any positive integer k , the quantity $E(X^k)$ is called the
 2 k^{th} moment for the distribution of X . We might call this the " k^{th} population
 3 moment" to be more clear.

4 Imagine that I have random variables $X_1, X_2, \dots, X_n$ that model the ob-
 5 servable sample. Then the k^{th} sample moment is

$$6 \quad M_k = \sum_{i=1}^n X_i^k / n$$

7 Note that M_k is our best estimator for $E(X^k)$, for instance $E(M_k) = E(X^k)$

$$E(M_k) = E\left(\frac{\sum X_i^k}{n}\right) = \frac{1}{n} n E(X_i^k) = E(X^k)$$

1 Now, suppose that $X_1, X_2, \dots, X_n$ are i.i.d. each with density $f_X(x; \theta)$.

2 Here, $\theta = (\theta_1, \theta_2, \dots, \theta_p)$ is a vector of parameters.

3 Set up a system of p equations, setting the first p sample moments equal to
 4 the first p population moments:

$$5 \quad M_1 = E(X)$$

$$6 \quad M_2 = E(X^2)$$

$$7 \quad \vdots \quad \vdots$$

$$8 \quad M_p = E(X^p)$$

9 can be observed functions of $\theta_1, \theta_2, \dots, \theta_p$

10 We note that the sample moments are all quantities we can determine from
 11 the available sample. We also note that the population moments are func-
 12 tions of the unknown parameters.

1 We can now solve this system of equations for $\theta_1, \theta_2, \dots, \theta_p$:

2
$$\theta_1 = g_1(X_1, X_2, \dots, X_n)$$

3
$$\theta_2 = g_2(X_1, X_2, \dots, X_n)$$

4
$$\vdots \quad \vdots$$

5
$$\theta_p = g_p(X_1, X_2, \dots, X_n)$$

7 These equations are built on the (incorrect) belief that $M_i = E(X^i)$.

8 In fact, $M_i \approx E(X^i)$, so

9
$$\theta_i \approx g_i(X_1, X_2, \dots, X_n)$$

10 and we construct our estimator as

11
$$\hat{\theta}_{i, \text{MOM}} = g_i(X_1, X_2, \dots, X_n)$$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. from the Exponential(λ)
 2 distribution. (Recall that this means that $E(X_i) = 1/\lambda$.) What is the method
 3 of moments estimator for λ ?

4 *p=1 here, need only one equation*
 $m_1 = \sum x_i/n = \bar{x}$

6 *$E(X) = 1/\lambda \leftarrow \text{first pop. moment}$*

8 *Set equal to each other $m_1 = E(X)$*
ie. $\bar{x} = 1/\lambda$

11 *Solve for parameter: $\lambda = 1/\bar{x}$*

12 *So, MOM estimator for λ is $\hat{\lambda}_{\text{mom}} = 1/\bar{x}$*

Note that $E(\hat{\lambda}_{\text{mom}}) \neq 1/E(\bar{x}) = \lambda$, so $\hat{\lambda}_{\text{mom}}$ is biased for λ